**3. State Sync & Audit Log:** Results write to Cosmos DB checkpoint with full trace context.

# 5. Low-Level Design (LLD) & Execution Sequence

The low-level design maps the FastAPI controllers, LangGraph state graph nodes, Pydantic data structures, and Azure Function async tool invocations into an execution flow sequence:

### LOW-LEVEL DESIGN (LLD): CLASS RELATIONSHIPS & EXECUTION SEQUENCE

**FastAPI Ingress Controller**

```
+ submit_application()
+ get_execution_status()
+ resume_human_approval()
```

**LangGraph LoanState**

```
+ applicant_id: str
+ credit_score: int
+ verified_income: float
+ risk_category: Enum
+ messages: List[BaseMessage]
```

**Azure Function Tools**

```
+ parse_bank_statement()
+ verify_identity_doc()
+ fetch_credit_bureau()
```

**RUNTIME EXECUTION SEQUENCE FLOW:**

**Step 1 (Ingress):** Client POSTs loan application payload → FastAPI Router parses & initializes `LoanState` schema.

**Step 2 (Delegation):** LangGraph Supervisor routes task via A2A protocol to `DocumentVerificationAgent`.

**Step 3 (Async Tool Execution):** Agent triggers Azure Function (OCR) via Blob Event grid → Updates state with extracted W2/Bank data.

**Step 4 (Underwriting & HITL):** `UnderwritingAgent` evaluates vector policy rules via Azure AI Search. If risk > threshold, triggers Human-in-the-Loop breakpoint.

**Step 5 (Checkpoint & Output):** Final state saved to Cosmos DB checkpoint storage → Returns A2A decision payload to API Gateway.